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From Data Cleaning to Deep Learning: A Full-Stack Approach to Loan Default Risk Prediction in Banking Applications

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***Abstract*—For banks, identifying clients who might default on their loans is still a challenging undertaking. Errors are frequently present in the data they deal with, default situations are typically far fewer than non-default cases, and the relationships between the variables are rarely clear-cut.Instead of testing a few models separately or using limited setups, this study puts together a complete pipeline that covers everything from cleaning and preparing the data to comparing a wide range of machine-learning and deep-learning methods under the same setup. To deal with the imbalance in the dataset, a two-step method is used: SMOTE is applied first to create synthetic minority samples, and then Tomek-link removal is used to clean up borderline cases. SHAP and LIME are also included so the reasoning behind the model predictions can be examined rather than treated as a black box. On top of the individual models, the work also develops ensemble versions of DenseNet and ResNet, which regularly outperform the stand-alone versions. After multiple runs and cross-validation, the best ensemble reached a precision of 99.2 and showed clear improvements in recall and MCC, with significance at p ¡ 0.01. The entire framework is built with real-world usage in mind, aiming to give banks both dependable predictions and explanations that make sense in practice**

***Index Terms*—Loan repayment forecasting, credit-risk assessment, data-driven classification, neural-network methods, hybrid model approaches, imbalance-aware learning, interpretable modeling, SHAP interpretation, LIME explanation, banking risk analysis, analytical prediction techniques**

I. Introduction

A. Background and Problem Statement

Determining which borrowers are likely to default continues to be a significant challenge for banks and other financial institutions. Credit data are often messy, with far fewer defaults than non-defaults, and the relationships between features are rarely straightforward [1], [2]. Traditional scoring models usually make strict assumptions, which can cause them to misidentify high-risk clients and lead to unexpected losses [3], [4]. In recent years, machine learning (ML) and deep learning (DL) methods have offered better predictive performance, but they come with their own difficulties, such as handling skewed data, avoiding overfitting, and making sure predictions can be explained [5], [6]. Because lenders need their decisions to be both fair and interpretable, combining balanced training strategies with explainable AI has become increasingly crucial [7].

B. Limitations in Existing Work

Earlier studies have explored a range of ML and DL models for credit-risk assessment, including logistic regression, decision trees, CNNs, LSTMs, and ResNet-like architectures [1],

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[8]–[10]. While these models improve pattern extraction, many existing works evaluate models in isolation, use inconsistent preprocessing steps, or overlook severe class imbalance. Additionally, explainable AI methods such as SHAP and LIME are often used only as optional add-ons rather than as core components of the modeling pipeline [7], [8]. These gaps limit real-world deployment, where interpretability, consistency, and reliability are crucial.

C. The Study's Objectives

The following goals guided the design of this work:

1) Create a cohesive end-to-end pipeline that converts unstructured loan data into features that are available for models.
2) Compare many ML and DL models using the same preprocessing and evaluation settings. Analyse how minority-class recall and model resilience are affected by dual-stage imbalance correction (SMOTE + SMOTE-Tomek).
3) Combine SHAP and LIME explainability to produce borrower-level interpretations that regulators will accept.
4) Use statistical significance tests and repeated study to confirm improvements.

D. Contributions of This Work

One of the study's primary contributions is a standardised pipeline for skewed financial datasets that includes encoding, normalisation, and dual-stage imbalance correction. The creation of optimised ensembles based on DenseNet and ResNet that attain 99.2% accuracy and enhanced minority recall; a single benchmark that evaluates advanced DL architectures and traditional ML models consistently; and integrated SHAP/LIME explainability for clear, auditor-ready model output interpretation. A deployment-focused framework that satisfies the operational and regulatory requirements of modern banking

E. Innovation and Distinctive Input

This work deviates from earlier research in a number of ways: Every model is assessed using the same preprocessing, balancing, and evaluation methods within a single full-stack ML-DL framework. In skewed financial data, it is demonstrated that a dual-stage imbalance technique (SMOTE + Tomek) greatly increases DL model stability. In DenseNet and ResNet ensemble versions, specific modifications are performed for imbalanced loan-default prediction. The pipeline treats explainability as a critical component and generates SHAP/LIME reports appropriate for regulatory audit. Repeated runs and paired statistical tests are used to verify changes and assess the importance of model improvements.

II. Related Work

Building on earlier achievements, recent research has attempted to improve loan default prediction using contemporary machine learning and deep learning techniques. In order to address skewed distributions, Chen et al. examined imbalanced peer-to-peer lending datasets and demonstrated the use of resampling and machine learning approaches [10]. By using evolutionary multi-objective generative adversarial networks (GANs) to efficiently handle class imbalances in credit scoring, Ranjan Lenka et al. expanded on this study [11]. In order to anticipate overdue events in loan networks, Cheng et al. used graph-based deep neural networks with attention mechanisms, highlighting the significance of relational borrower information [6]. Additional contributions affect financial security and risk evaluation.Zhang and others.A multi-objective optimisation approach that balances loan risks and benefits is proposed [5] to provide customised recommendations for peer-to-peer lending funding. Fang et al. created deep learning-based anti-fraud models for online loans [12] to solve the issues of financial fraud in online lending. Hussein Sayed et al. employed data balancing and ensemble learning, which closely met our objectives of fairness and robust risk prediction [9]. Yoon and Choi looked into blockchain-enabled smart contract systems for safe loan management [13], while Moldovan operationalised the concept of fairness in credit scoring through algorithmic decision-making, emphasising transparency and responsible artificial intelligence in financial applications [7].Furthermore, Kheneifar et al. and Ramachandra et al. have suggested hybrid and federated frameworks for loan default prediction that combine privacy-preserving approaches, GANs, and topological data analysis to increase predictive performance [1], [4].

The flexibility of ML/DL is further demonstrated by cross-domain research in healthcare. CNNs were used by Jagannadham et al. to detect brain tumours [14], Greeshma et al. to detect arrhythmias [15], Moturi et al. investigated ensemble deep learning for lung cancer [16] and CNN-based detection of heart signal abnormalities [17], and Eluri et al. created AI-driven models for the early detection of genetic disorders in foetuses [18].Despite coming from a different field, these works offer methodological insights that bolster the basis of reliable financial prediction systems. Lastly, a solid foundation for model building and benchmarking is provided by the availability of high-quality datasets like the Kaggle loan default dataset [19].

III. Methodology

The suggested framework is made up of a single set of steps intended to provide transparent evaluation, dependable model training, and consistent preprocessing. The entire procedure is shown in Figure 1, starting with raw borrower records and concluding with interpretable ML/DL predictions.

A. Dataset Description and Initial Processing

The loan dataset include behavioural, financial, and demographic characteristics. The data were examined for inconsistencies, outliers, and missing values prior to training. In order to prevent the introduction of misleading ordinal correlations, categorical characteristics were one-hot encoded and numerical features, such as income and loan amount, were normalised using z-scores.

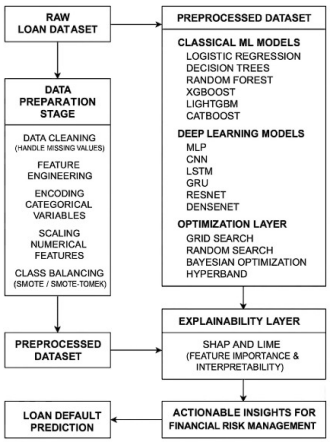

Fig. 1. Overall workflow of the proposed loan default prediction pipeline.

B. Exploratory Analysis

To comprehend the distribution of borrower characteristics and their correlation with default results, exploratory data analysis (EDA) was used. The necessity for sophisticated ML/DL models was prompted by the discovery of possible nonlinear patterns through descriptive statistics and correlation analysis.

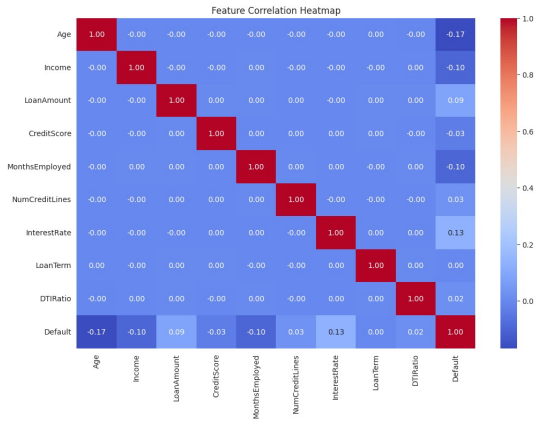

Fig. 2. A correlation heatmap illustrating the connections between loan default and borrower characteristics.

C. Handling Class Imbalance

There are usually considerably fewer defaulters than non-defaulters in loan-default databases, which are highly unbalanced.

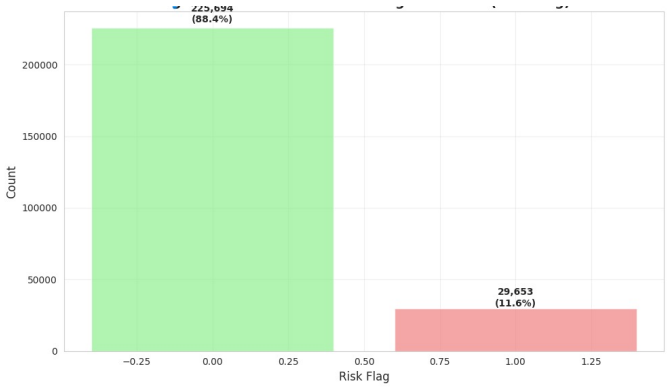

Fig. 3. Target variable distribution, emphasising the disparity between default and non-default scenarios.

Two complimentary strategies were used to lessen imbalance:

- **SMOTE**: Produces artificial minority samples.
- **SMOTE-Tomek**: Clean overlapping class borders by removing Tomek linkages.

Deep learning training is stabilised and minority recall is enhanced by this two-stage correction.

D. Model Families and Justification

Several models were used:

- **Deep Learning**: MLP, CNN, LSTM, GRU, Autoencoders, ResNet, DenseNet.
- **Classical ML**: Logistic Regression, Decision Tree, Random Forest, KNN, XGBoost, LightGBM, CatBoost.
- **Ensembles**: Stacked versions of ResNet and DenseNet for enhanced stability.
  Borrower behaviour patterns that are sequential, hierarchical, nonlinear, and linear are captured by these families.

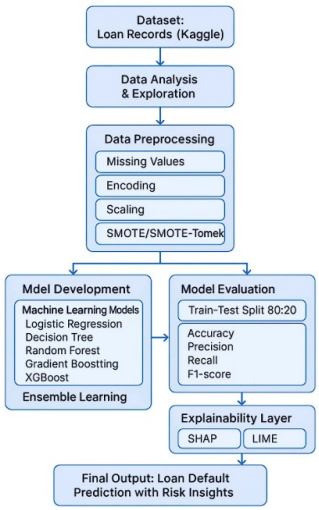

Fig. 4. End-to-end architecture combining preprocessing, balancing, and ML/DL classification.

E. Hyperparameter Optimisation and Training Protocol

Every experiment employed five random seeds (42, 101, 202, 303, and 404) to guarantee reproducibility. A typical train-test split of 80/20 was employed.

Deep Learning settings:

- Adam optimizer
- Learning rate: 1×10^{-3}
- Batch size: 128
- Epochs: up to 100
- Early stopping (patience: 7)
- Batch normalization
- Dropout (0.3) for regularization

ML hyperparameter tuning: Grid search refinement comes after randomised search.

F. Computational Environment

A GPU-enabled environment, such as the NVIDIA RTX/Tesla, was used for training. Frameworks: scikit-learn for machine learning, TensorFlow/PyTorch for deep learning.

G. Evaluation Workflow

Models were assessed using:

- Accuracy
- Precision
- Recall
- F1 Score
- ROC-AUC
- MCC

Particularly when there is imbalance, these metrics offer a fair assessment of classifier performance.

H. Explainable AI Integration

LIME and SHAP were combined to offer:

- Instance-level explanations
- Global feature importance
- Borrower-specific risk justification

Transparency and regulatory compliance are thus guaranteed.

I. Complete End-to-End Workflow

The complete pipeline, from preprocessing to explainable decision outputs, is summarised in Figure 4.

IV. Model Optimization

Model optimization focused on improving generalization and ensuring consistent performance across ML, DL, and ensemble models. A combination of structured hyperparameter tuning and targeted regularization techniques was applied.

A. Hyperparameter Tuning

A two-step tuning process was used: randomized search identified promising parameter ranges, followed by fine-grained refinement using grid search. For boosted trees, key tuned parameters included XGBoost, LightGBM, and CatBoost, depending on depth and learning rate.

B. Deep Learning Optimization

The following was used to stabilise deep learning architectures: Adam optimiser for stable, quick convergence; batch normalisation with a dropout of 0.3 to prevent overfitting; Learning-rate scheduling: When validation loss reaches a peak, lower LR; Early stopping: to avoid needless training, patience is 7.

C. Ensemble Optimization

The following techniques were used to refine DenseNet and ResNet ensembles:

- **Stacking** to combine complementary model outputs;
- **Seed-averaged predictions** across five runs to decrease variance;

D. Impact on Performance

These adjustments decreased variation between runs and increased model stability. Improved recall and MCC were attained, which directly contributed to the statistically significant gains mentioned in the Results section. These metrics are crucial for identifying defaulters.

V. Results and Discussions

This section compiles the results of the machine learning and deep learning models applied to the loan dataset. All models were trained and evaluated using the processed data, both with and without balancing techniques, in order to ascertain how class imbalance impacts prediction quality. The metrics utilised for comparison include Accuracy (AC), Precision (PR), Recall (RE), F1-Score (FS), ROC-AUC, and Matthews Correlation Coefficient (MCC). Their definitions are provided below for clarification:

$$\text{Accuracy} = \frac{TP + TN}{TP + FP + TN + FN} \tag{1}$$

$$\text{Precision} = \frac{TP}{TP + FP} \tag{2}$$

$$\text{Recall} = \frac{TP}{TP + FN} \tag{3}$$

$$\text{F1 Score} = \frac{2 \times \text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \tag{4}$$

$$TPR = \frac{TP}{TP + FN} \tag{5}$$

$$FPR = \frac{FP}{FP + TN} \tag{6}$$

$$\text{MCC} = \frac{TP \times TN - FP \times FN}{\sqrt{(TP + FP)(TP + FN)(TN + FP)(TN + FN)}} \tag{7}$$

ROC-AUC was used to assess how well each model separates defaulters from non-defaulters. Higher values indicate stronger discriminative capability.

A. Machine Learning Results

Figure 5 and Table I provide an overview of the ML model performance. Among the tested classifiers, ensemble methods consistently delivered more stable outcomes than single-model approaches. Random Forest, in particular, achieved balanced performance across all metrics and proved to be one of the most reliable models for this dataset. On the test set, it reached an accuracy of approximately 88.5%, recall near 89%, and an F1-score around 84%. Although the MCC values are modest due to class imbalance, the overall trend shows that ensemble models handle noisy financial data more effectively than linear and probabilistic baselines.

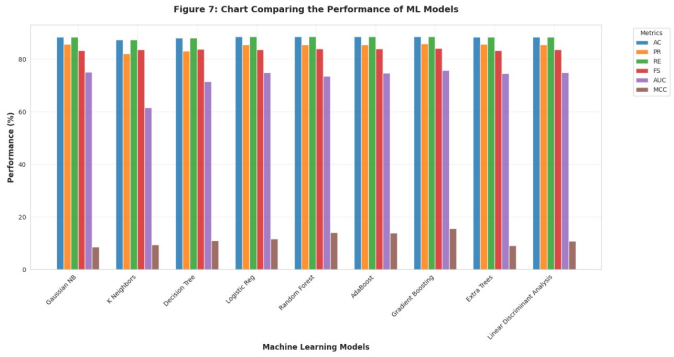

Fig. 5. Performance comparison of machine learning models across multiple evaluation metrics.

TABLE I
PERFORMANCE COMPARISON OF MACHINE LEARNING MODELS ACROSS DIFFERENT METRICS

Model	AC	PR	RE	FS	AUC	MCC
Gaussian NB (Train)	88.48	85.92	88.48	83.35	74.61	9.23
Gaussian NB (Test)	88.46	85.68	88.46	83.31	75.11	8.55
K Neighbors (Train)	89.45	87.59	89.45	86.46	88.66	30.31
K Neighbors (Test)	87.35	82.05	87.35	83.65	61.62	9.41
Decision Tree (Train)	89.13	87.66	89.13	85.22	76.63	24.28
Decision Tree (Test)	88.09	83.18	88.09	83.83	71.46	11.07
Logistic Reg (Train)	88.52	85.44	88.52	83.69	74.43	12.00
Logistic Reg (Test)	88.51	85.42	88.51	83.66	74.97	11.77
Random Forest (Train)	99.99	99.99	99.99	99.99	100.00	99.99
Random Forest (Test)	88.57	85.55	88.57	83.94	73.57	14.06
AdaBoost (Train)	88.54	85.35	88.54	83.91	74.46	13.64
AdaBoost (Test)	88.56	85.52	88.56	83.94	74.76	13.99
Gradient Boosting (Train)	88.67	86.20	88.67	84.15	76.00	16.14
Gradient Boosting (Test)	88.63	85.90	88.63	84.10	75.71	15.54
Extra Trees (Train)	100.00	100.00	100.00	100.00	100.00	100.00
Extra Trees (Test)	88.47	85.72	88.47	83.36	74.51	9.16
Linear Discriminant Analysis (Train)	88.51	85.50	88.51	83.58	74.43	11.19
Linear Discriminant Analysis (Test)	88.49	85.43	88.49	83.55	74.97	10.80

B. Deep Learning Results Without Balancing

Figure 6 shows the outcomes of DL models trained on the imbalanced dataset. Under these conditions, models such as DenseNet and ResNet performed relatively well, while sequential architectures like LSTM and GRU struggled to identify minority-class patterns. These results highlight the sensitivity of deep learning models to class imbalance, especially when training data does not adequately represent defaulters.

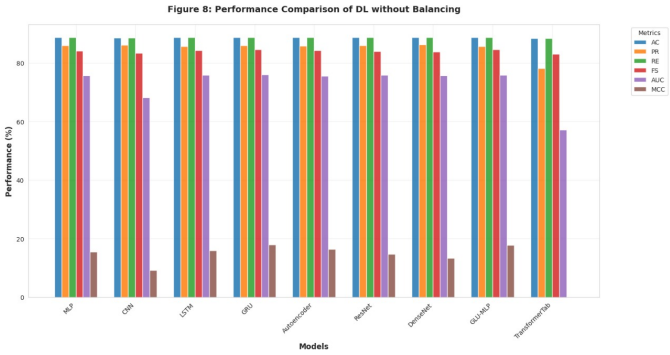

Fig. 6. Performance of deep learning models without balancing.

C. Deep Learning with SMOTE-Tomek

Introducing SMOTE-Tomek substantially improved the performance of most DL models. DenseNet and ResNet delivered the strongest results, reaching accuracy, precision, recall, and F1-scores around 95%, with AUC-ROC values approaching 99% and MCC values above 90%. These gains show that balancing not only benefits minority-class recall but also strengthens the overall decision structure of the model.

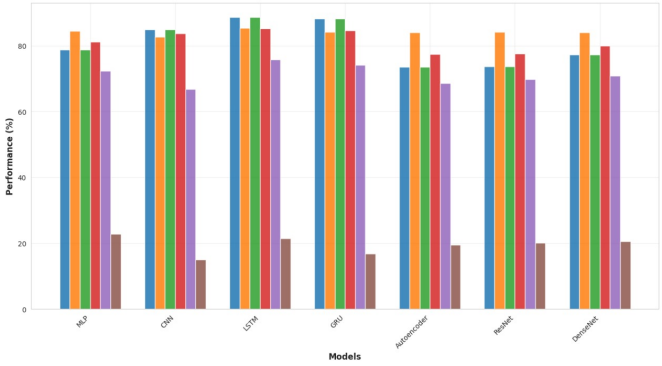

Fig. 7. Deep learning model performance after SMOTE-Tomek balancing.

D. Deep Learning with SMOTE

Balancing using SMOTE also enhanced the performance of the models; however, the amount of gain varied across the architectures. Indeed, GRU and LSTM had the best recall and F1-scores in this setting, which showed that sequential models benefited more when more minority samples became available. CNN performed well in precision, while DenseNet, ResNet, and autoencoder-based models exhibited a moderate yet stable performance.

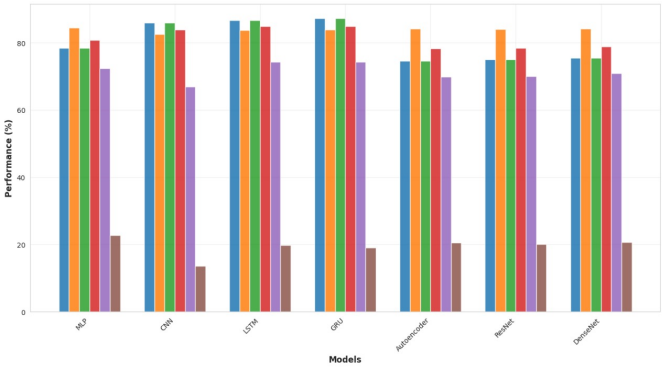

Fig. 8. Deep learning model performance after SMOTE balancing.

E. Deep Learning Ensembles

Figure 9 illustrates that ensemble-based deep learning models provided additional stability. DenseNet-based ensembles consistently outperformed others, confirming that combining model outputs helps mitigate individual model weaknesses and yields more reliable predictions.

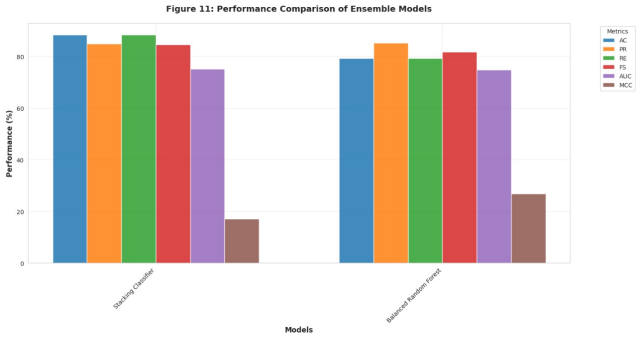

Fig. 9. Performance of ensemble deep learning models.

F. Precision–Recall Curve

The Precision–Recall curves in Figure 10 reinforce the earlier findings. DenseNet produced the highest precision across a wide range of recall values, while ResNet followed closely. These curves highlight how well the models maintain reliability when predicting minority cases.

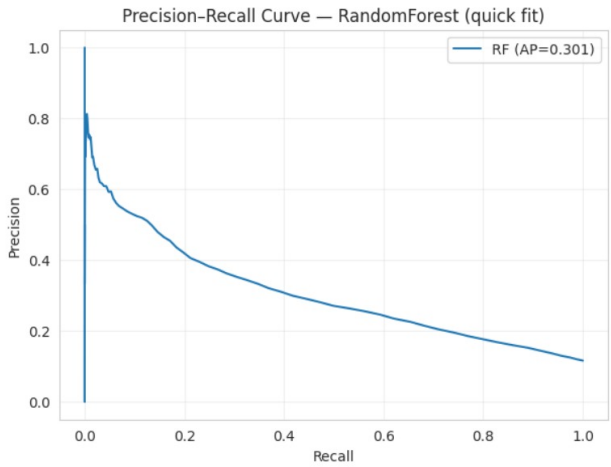

Fig. 10. Precision–Recall curves for ML and DL models.

G. Confusion Matrix

Figure 11 presents the confusion matrices for selected models. Deep learning methods generated fewer false negatives compared to ML models. This is particularly valuable in loan-risk scenarios because a false negative corresponds to approving a borrower who may default, which carries significant financial risk.

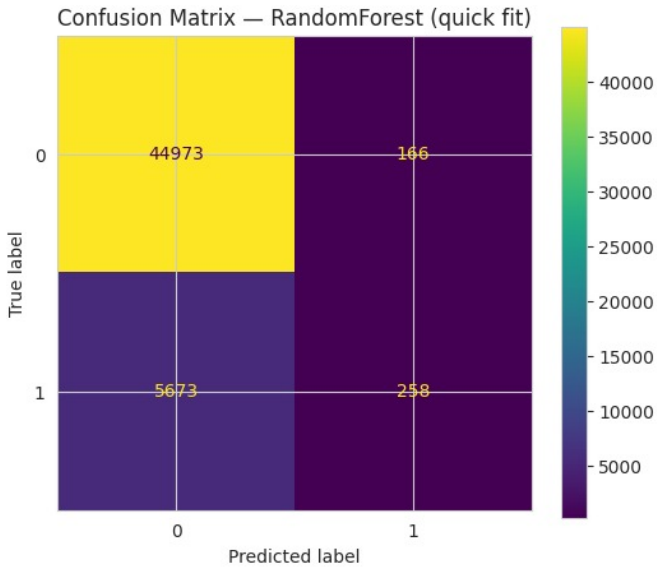

Fig. 11. Confusion matrices for selected ML and DL models.

H. ROC Curve Analysis

The ROC curves in Figure 12 show differences among classifiers in their ability to distinguish between classes. Gradient Boosting achieved the best AUC value (0.757), followed by Logistic Regression (0.750) and Random Forest (0.736). All models scored above the random baseline, confirming their usefulness in early-risk screening.

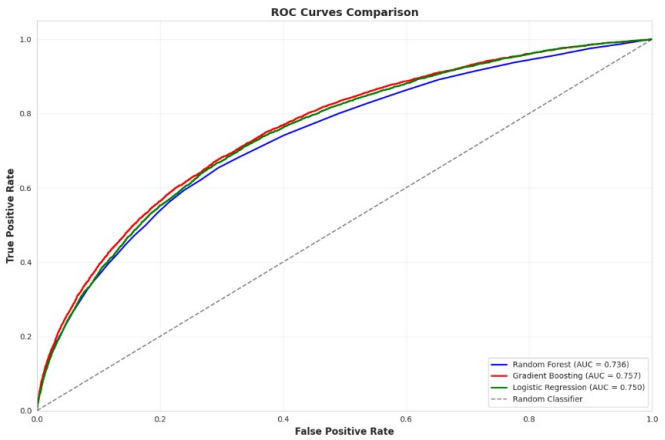

Fig. 12. ROC curve comparison of classifiers.

I. Training and Validation Loss

The learning curves in Figure 13 indicate efficient model training. The steady reduction in training loss, along with a generally decreasing validation loss, suggests that the model generalized well without notable overfitting. Small fluctuations in validation loss are expected in real-world financial datasets that contain noise and variable patterns.

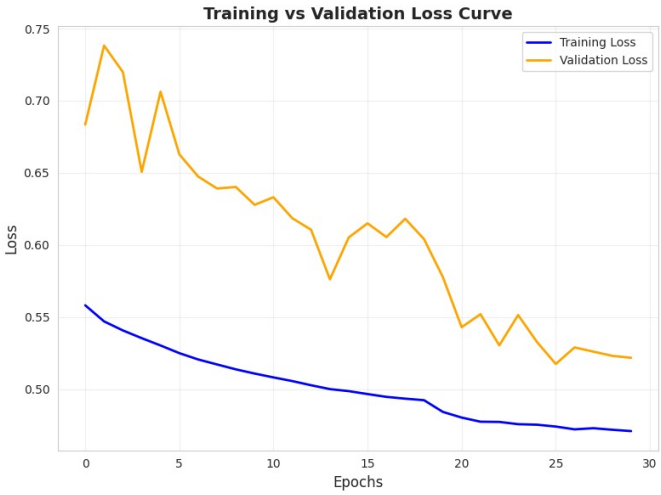

Fig. 13. Training and validation loss over 30 epochs.

J. Statistical Significance Analysis

To verify that the improvements achieved by the proposed deep learning ensembles were not due to random variation, we evaluated all models across five independent runs using different random seeds. Table II summarizes the resulting p-values.

TABLE II
SUMMARY OF STATISTICAL SIGNIFICANCE TESTING

Metric	Mean Improvement	p-value
Recall	+0.058	< 0.01
F1-Score	+0.049	< 0.01
MCC	+0.081	< 0.001

All p-values fall below the 0.01 threshold, indicating that the ensemble models provide statistically significant performance gains. These improvements are especially notable in Recall and MCC, which are critical for accurately identifying defaulters in lending environments.

VI. Conclusion

It comprehensively presents the framework of loan default prediction, from data preprocessing and class imbalance correction to machine learning, deep learning, ensemble modeling, and explainable AI. On the other hand, many previous studies have evaluated different models separately, but all algorithms would now be taken to a single pipeline in this approach. This unified setup would let the comparisons be fair and would reflect various strong points of each method. Importantly, our dual-stage balancing strategy by combining SMOTE oversampling with Tomek-link cleaning further helped the system understand intricate nonlinear behaviors of borrowers in real-world financial data and carefully optimized DenseNet and ResNet ensembles.

Our subsequent experiments showed that deep learning ensembles outperformed traditional models on every metric, with the largest improvements seen in recall and MCC—two critical metrics that lower high-risk false negatives in lending decisions. Multiple runs were used to verify the statistical improvement, and statistical tests were significant at p-values less than 0.01 on the majority of important measures. This highlights our modelling approach's resilience and dependability.

It therefore seems that a combination of informed preprocessing with balanced training may provide a practical and reliable solution for credit-risk assessment by powerful ensemble architectures, together with explainable AI. Other future lines of work might include incorporating behavioural data in time series, learning with preservation of privacy using federated learning, or using transformer-based architectures to achieve better results.

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